SHIP TYPES 1.3 · 10 MIN

Container Ships

The box, size classes, design drivers, feeders and our two Baseline ships · Session W2.3 · Not in the Evaluation Test

Tiago Santos, Luís Batista

Instituto Superior Técnico / University of Lisbon

tiago.santos@centec.tecnico.ulisboa.pt

luis@veranavis.com

A Ship Built Around a Standard Box

ISO 668 type	Length	Width	Height	Max gross
1CC · 20 ft, the TEU	6.058 m	2.438 m	2.591 m (8′6″)	36,000 kg
1AA · 40 ft, the FEU	12.192 m	2.438 m	2.591 m	36,000 kg
1AAA · 40 ft high cube	12.192 m	2.438 m	2.896 m (9′6″)	36,000 kg

Below deck

Cell guides hold each stack: no lashing.

On deck

On hatch covers, held by twistlocks and lashing rods.

Every slot has an address

Bay (along), row (across), tier (up).

Bays set L, rows set B, tiers set D: the hull steps in units of the box.

From Feeders to 24,000 TEU

Class	Capacity	What limits it
Small feeders, coastal	up to about 1,000 TEU	Port depth and berth length; often geared
Feeders, short sea	about 1,000–3,000 TEU	Regional ports; hub-to-port service
Panamax (old locks)	up to about 5,000 TEU	B 32.31 m, T 12.04 m, LOA 294 m
Neopanamax	up to about 13,000–17,000 TEU	B 51.25 m, T 15.2 m, LOA 366 m
Ultra-large (ULCV)	today up to 24,346 TEU	Ports, cranes, Suez; about 400 × 61 m

5,736 container ships in 2022, 280 million GT (Equasis)

Fleet capacity over **33 million TEU** in November 2025 (Alphaliner)

What Shapes a Container Ship

Volume, not weight

Slots fill before the load line: excess freeboard, many tiers on deck.

Speed

Liner schedules: fine hulls, CB about 0.6–0.7, Fn about 0.2–0.3.

Stability with deck stacks

High KG and wind area: beam and ballast keep GM up. Sensitive to parametric roll.

The view ahead

SOLAS V/22 limits how far the deck stacks may hide the sea: bridge position, stack height forward.

Open holds, wide hatches: torsion and hull girder strength are a design issue on their own.

Gearless or Geared: Who Brings the Crane?

Gearless · the Express

The port’s gantry cranes work the ship: fast, and no crane weight on board.

Needs ports with gantries on every call.

Geared · the Feeder

Own crane on deck: reaches ports with no gantry, at a lower handling rate.

Costs weight high up, deck space and a heel while lifting.

Big ships call at a few deep hubs; feeders carry the boxes on. Our route is a small network: the **Express** links Lisbon to the two main islands, the **Feeder** serves six more.

> For the Feeder the crane is a design input: safe working load, outreach to the far row, and stability with a box on the hook.

How Heavy Is a TEU?

Nominal TEU

Every slot filled, whatever the weight. Counts the space.

TEU at 14 t

Boxes of 14 t each, homogeneous: the industry’s check that deadweight and stability can carry them.

Fleet average

Appendix A: payload about 10 t per TEU, and payload = 0.75 DWT.

> 10 t or 14 t per TEU moves the deadweight of the Express by 40%. Our brief fixes 10 t on average: 1,300 TEU is 13,000 t of cargo.

Reefers add weight and a power plug each; empties go back light. Island trades carry both.

Where the Express and the Feeder Sit

Azores Express · 1,300 TEU

A gearless feeder on a weekly liner service: ports with gantry cranes, fine hull for 18–19 kn, forward bridge.

Inside the container data of Appendix A.

Azores Feeder · 130 TEU

A small geared coastal ship, its own crane for island ports, batteries on deck. Hull suggested: PSV type. Nearest built ship: Yara Birkeland, 120 TEU, battery-electric, but 6 kn on a 7 nm route.

At the edge of the container data: look further.

Question for your project (D2): which family of Appendix A describes your ship best, and why?